Inverse of multivector: Beyond p+q=5 threshold

Arturas Acus

Vilnius university, institute of theoretical physics and astronomy
Sauletekio 3, room A409
Lithuania
arturas.acus@tfai.vu.lt

Joint work with: A. Dargys

The algorithm of finding inverse multivectors (MV) in a symbolic form is of paramount importance in computational Clifford or geometric algebra (GA). First attempts of inversion of general MV (J.P. Fletcher 2002) were based on decomposition of MV into basis elements. The complexity of a such calculations, however, grows exponentially with the dimension $n = p + q$ of $Cl_{p,q}$ algebra. The breakthrough occurred 10 years later (P. Dadbeh 2011), after arbitrary grade-negation operation for MVs was introduced. This allowed to write down compact and explicit MV inverse formulas as a product of initial MV and its carefully chosen grade-negation counterparts for all GAs up to dimension $n \leq 5$.

It this report we show that the grade-negation method can be extended beyond $p + q = 5$ threshold if, in addition, properly constructed linear combinations of grade-negated MVs are used. In particular, we write down explicit compact MV inverse formulas for $p + q = 6$ algebra.

The derivation of the inverse formulas was done with *Mathematica* package `https://github.com/ArturasAcus/GeometricAlgebra` for symbolic GA calculations, which can also be found in *Mathematica* community packages data base (`http://packagedata.net`). It currently implements the following features: textbook notation and precedences of products (without use of *Mathematica* palettes); algebraic operations in orthonormal frame (additive representation, symbolic coefficients); matrix representations of general $Cl_{p,q}$ algebras; easy switching between multiple algebras during the same Mathematica session; calculation of idempotents with different base element sorting; main involutions and general multivector inverse.

The package distribution also includes a number of test notebooks with a lot of algebraic formula tests, examples of generated matrix representations, tables of idempotents, etc. Short demonstration of some of these features as well as essential steps of inverse formula derivation process for $p + q = 6$ algebra will hopefully be interactively presented during the presentation.